

Person recognition

1. First determine whether the version of the image transmission module is the AI version firmware. **If you already know the firmware version, you can ignore this step** and use the **wifi_ver** command to query, as shown in the figure below

```
[2024-08-06 14:53:29.948]# SEND ASCII>
wifi_ver
[2024-08-06 14:53:30.010]# RECV ASCII>
YAHBOOM VerSion:AI_V0.5.0
```

2. Switch the mode to face recognition mode. Use the following command to send **ai_mode:4** through the serial assistant. After successful configuration, the module will restart and print the current firmware version, and then face recognition can start, as shown in the figure

```
[2024-08-06 15:03:04.205]# SEND ASCII>
ai_mode:4
[2024-08-06 15:03:04.341]# RECV ASCII>
ESP-ROM:esp32s3-20210327
Build:Mar 27 2021
rst:0x3 (RTC_SW_SYS_RST),boot:0x8 (SPI_FAST_FLASH_BOOT)
Saved PC:0x40375a0c
SPIWP:0xee
mode:DIO, clock div:1
load:0x3fce3818,len:0x514
load:0x403c9700,len:0x4
load:0x403c9704,len:0xb58
load:0x403cc700,len:0x2a8c
entry 0x403c98fc

[2024-08-06 15:03:04.886]# RECV ASCII>
[0:32mI (549) cpu_start: Multicore app[0m
[0:32mI (549) esp_psram: Found 2MB PSRAM device[0m
[0:32mI (550) esp_psram: Speed: 80MHz[0m
[0:32mI (550) cpu_start: Pro cpu up. [0m
[0:32mI (554) cpu_start: Starting app cpu, €
[2024-08-06 15:03:07.070]# RECV ASCII>
YAHBOOM AI VerSion:AI_V0.5.0
```

Note: If the current mode is already face recognition mode, the module only returns ok

```
[2024-08-06 15:05:32.444]# SEND ASCII>
ai_mode:4
[2024-08-06 15:05:32.503]# RECV ASCII>
OK
```

3. Send **KEY_MENU** through the serial port to enter the currently detected face. If the entry is successful, the image will be displayed as shown below

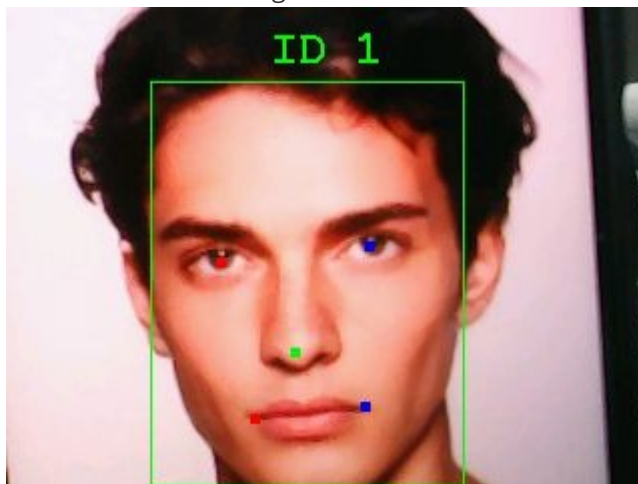


4. Send **KEY_PLAY** through the serial port to make the module start recognizing the current face and send the coordinates of the upper left and lower right corners of the first frame and the entered face number through the serial port.

The specific content is **\$upper left corner x-axis coordinate, upper left corner y-axis coordinate, lower right corner x-axis coordinate, lower right corner y-axis coordinate, # @ID: face number!**

Note: You need to open the screen to start recognition. If the screen is closed in the middle, the recognition will also be closed. If the face is not entered, the face number is -1

Entered face and recognized



Not entered face and detected



Serial port information:

```
KEY_PLAY OK
$71, 39, 228, 253, #@ID:1!
KEY_PLAY OK
$82, 6, 236, 229, #@ID:-1!
```

5. If you want to delete the face entered into the system (starting from the last entry), send **KEY_DOWN** through the serial port to delete it. **You need to detect faces before you can delete them. If you delete successfully, the corresponding result "X IDS left" (x represents the number of faces still stored in the system) will be displayed, as shown in the figure**

